Akshit Harsola

MACHINE LEARNING ENGINEER | ROBOTICS SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

Intelligent systems engineer specialising in machine learning and robotics: imitation-learning for manipulation, perception, embedded vision and production CV pipelines. MSc in Intelligent Robotics (University of Galway, 2:1) with three published papers across computer vision, distributed systems and computational biology. Delivered client work in mobile development, sensor data integration and workflow automation. Seeking graduate and internship roles in Robotics, Computer Vision or Machine Learning in Ireland and the EU.

PROFESSIONAL EXPERIENCE

Freelance Software Developer

Independent Client Engagements

Confidential

2024 - 2025

- Diagnosed absent sensor data behind a client Power BI request, redirecting scope to data acquisition
- Constructed C++ and Python acquisition layer, supporting legacy sensors, exposing readings through an API
- Connected Power BI dashboards to the API with live updates and automated failure alerts
- Automated LaTeX formatting of client research papers, cutting a 2+ day task to 5-6 hours
- Developed Kotlin Android and Swift/SwiftUI iOS apps for several clients; shipped two production apps

PROJECTS

Imitation-Learning Policies for Robot Manipulation - MSc Major Project

Compared ACT, Diffusion Policy and SmolVLA imitation-learning policies on a SO-101 arm; final SmolVLA model succeeded in **8/10** pick-and-place trials (pilot scale)

Moved from a defective AL5D arm to the SO-101 after hardware faults made positioning unreliable

SO-101, ACT, Diffusion Policy, SmolVLA

Classical Parking Detection & Occupancy Classification

Designed a three-tier classical CV pipeline (Python/OpenCV) for parking occupancy on aerial imagery: 95.1% of 345 spots detected (328), 74.2% occupancy F1 (100% on easy images, 52.7% on medium), ~120 ms/image (9 fps)

Python, OpenCV, DBSCAN, Hough

Gazebo Mapping and Navigation Simulation - Sensing & Perception Course Project (EE5110)

Built a Gazebo simulation with LiDAR mapping and obstacle-avoiding navigation

Gazebo, ROS2, LiDAR

AutoVision - Automated Object Annotation System

Built a Python/PyQt5 tool for semi-automated image annotation (Canny edge detection, preprocessing, quality checks), cutting annotation time by **60%**

Python, PyQt5, Canny

iOS Credential Manager with Biometric Authentication

Built a Swift password manager with biometric authentication and encrypted backup/restore

Swift

EDUCATION

MSc in Intelligent Robotics

University of Galway

2025 - 2026

Result: 66.56%, Second Class Honours Grade 1 (2:1). Coursework: Machine Learning for Engineering, Multi-Agent Systems & Reinforcement Learning, Digital Twins for Robotics, Sensing & Perception for Autonomous Vehicles, Embedded Machine Vision, IoT Systems Design, Robotic Control & Navigation

B.Tech. in Computer Science

Medi-Caps University

2021 - 2025

CGPA: 7.74/10

RESEARCH PUBLICATIONS

Micro-Expression Recognition for Lie Detection Using Image Processing - International Journal of Science, Engineering and Technology (Apr 2025)

Integrating Blockchain and 5G Technologies for Enhanced Edge Computing: Opportunities, Challenges, and Solutions - International Journal of Recent Advances in Multidisciplinary Topics (Feb 2024)

Utilizing AlphaFold Predictions to Understand Protein Function and Dynamics - International Research Journal of Modernization in Engineering Technology and Science (Jan 2024)

TECHNICAL SKILLS

Machine Learning: Python, PyTorch, scikit-learn, Deep Learning, Model Optimisation, Reinforcement Learning

Data & Analytics: SQL, Pandas, NumPy, R/R Studio, Power BI (live dashboards, failure alerting), API Development

Robotics: ROS2 (Jazzy), Imitation Learning, Robot Manipulation, Digital Twins

Computer Vision: OpenCV, Object Detection, Embedded Vision, Real-time Vision Systems

Software & Tools: C/C++, Git, Docker, Linux/Ubuntu, CI/CD, Gazebo, LaTeX, Kotlin, Swift